Data-Driven Customer Reactivation Strategy

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Project Overview

The Problem

A significant segment of a company's Italian customer base has made no purchase in over 2 years, representing a dormant but potentially recoverable revenue stream.

The Goal

Predict which inactive clients have the highest propensity to reactivate, enabling a targeted campaign that maximizes ROI versus a blanket mass-mailing approach.

Available Data

5 years of transaction records (Client ID, date, order value, item type, tool flag) combined with firmographic data: sector, region, employees, economic potential, risk class.

Expected Outcome

A ranked, scored list of all 13,244 inactive clients ready for the marketing team with three actionable campaign tiers built in.

Data Understanding

13,244
Inactive Clients

5 yrs
Transaction History

875
Confirmed Reactivations

6.61%
Base Reactivation Rate

Transactional Data

- Individual purchase records across 5 years (2017–2021)
- Fields: Client ID, transaction date, NET order value, item type & tool flag
- Clear seasonality: Q4 spikes, summer dips, COVID-19 visible in early 2020
- Only purchase with positive NET values within 6 months accepted as valid reactivation signals (returns excluded)

Firmographic Data

- Region: Italian province codes enabling geographic pattern detection
- Trade Sector: Industry classification
- N_EMPLOYEES: Company size, proxy for purchase capacity
- ECONOMIC_POT: Total addressable market benchmark per client
- RISK_CAT: Credit/financial standing, turned out to be the #1 driver

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Who are our Inactive Clients?

12,369
Did NOT Reactivate
(93.39%)

875
Reactivated
(6.61%)

The Data Gap

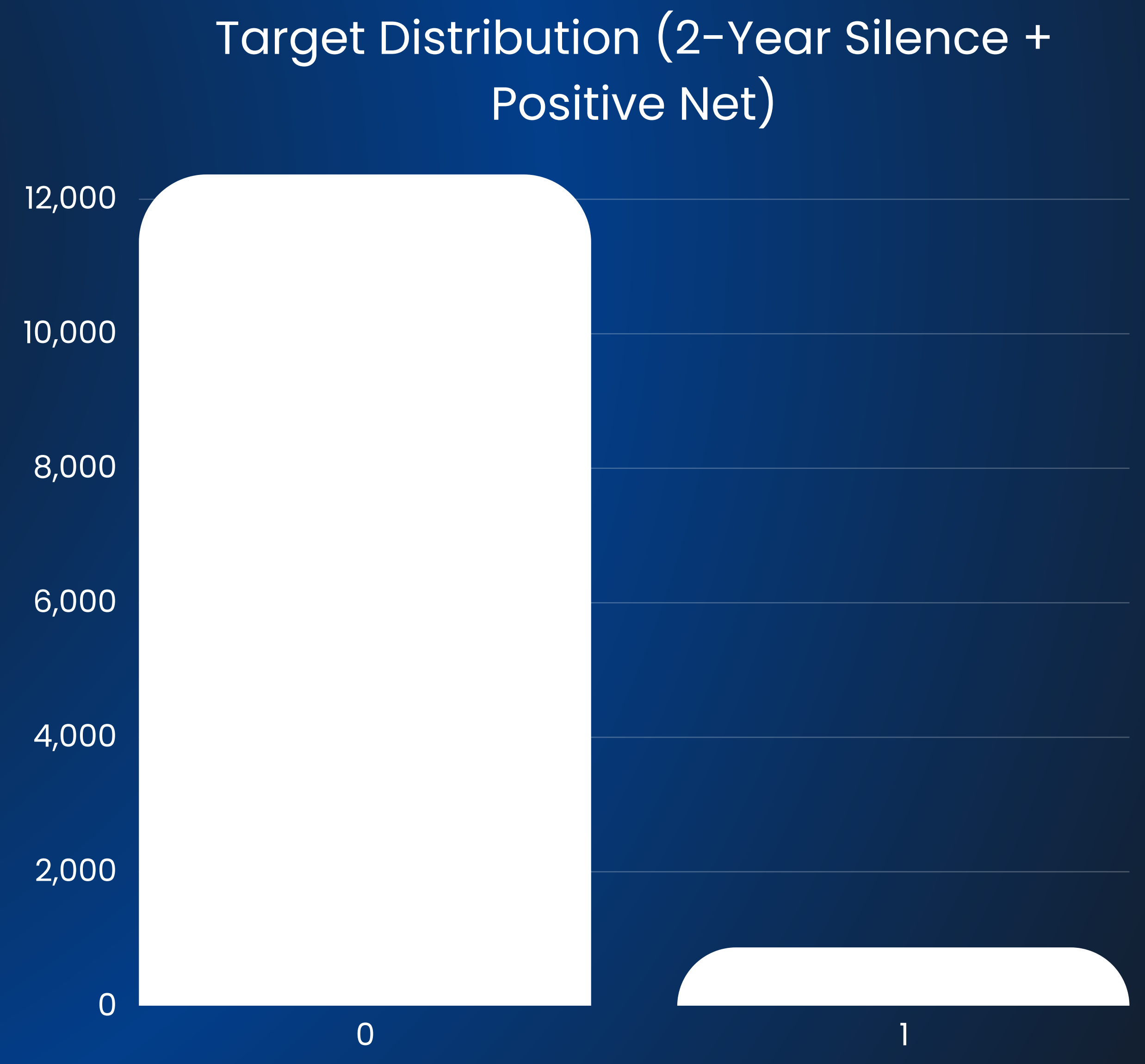
Only 875 of 13,244 inactive clients actually returned (6.61%). Most will never come back; the data is highly imbalanced.

Our Solution: Undersampling

We balanced the training set by undersampling the majority class at a 1:10 ratio, allowing the model to clearly learn the rare 'return' signal.

Result

The model became sensitive to hidden loyalty signals; patterns of past behaviour that distinguish the few who return from the many who don't.



Methodology

Training Window
2017 – 2019

45 client features are engineered from all purchase history. No future data enters, the model only knows what happened before Dec 31, 2019.

Observation Date
31 Dec 2019

The world freezes here. Every feature the model receives is computed up to this single point only. This eliminates data leakage entirely.

Target Window
Jan – Jun 2020

Did the client make at least one positive NET purchase in this 6-month window?
Yes → reactivated (1).
No → not reactivated (0).

Scoring
All 13,244 clients

The trained model assigns a reactivation probability (0–1) to every inactive client. Higher score = higher campaign priority.

Feature Engineering: 45 Signals of Intent

We didn't just look at their last purchase date. We created 45 unique indicators for each client, including:

Recency & Frequency

When did they last buy? How often? How many distinct products? Classic signals that separate occasional buyers from loyal ones.

Spending Patterns

Total historical spend, average order size, spend in the last 12 and 36 months. Are they a big spender or a small one?

Product Mix

Do they buy only tools, only accessories, or a mix of both? Clients with diverse purchasing habits behave differently.

Purchase Rhythm

How regularly do they buy? What is the average gap between purchases? Irregular buyers are different from seasonal ones.

Spending Trend

Was their spending growing or declining before they went silent? A client who was accelerating before stopping is different from one who was already fading.

Company Profile

Sector, region, size, economic potential. B2B clients in certain sectors or regions may have industry-specific reactivation patterns.

The Engine - All evaluated models

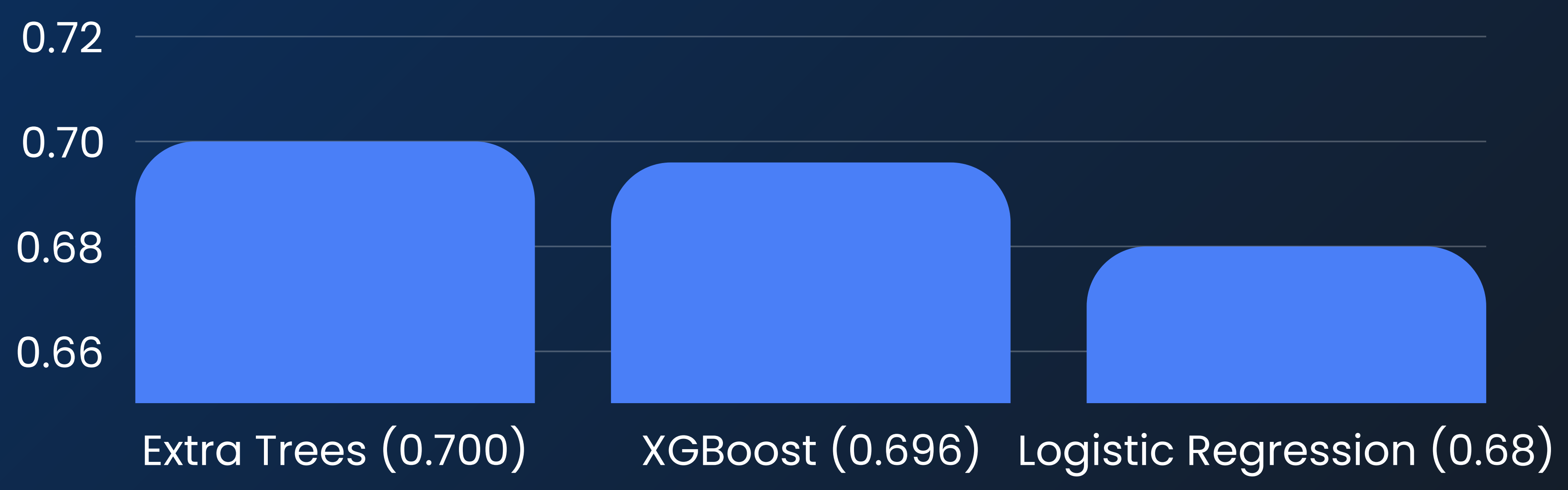
Why Extra Trees?

Ensemble of randomised decision trees
Extra Trees builds hundreds of trees, each trained on a random feature subset with fully randomised split points — making it highly robust to noisy or correlated features.

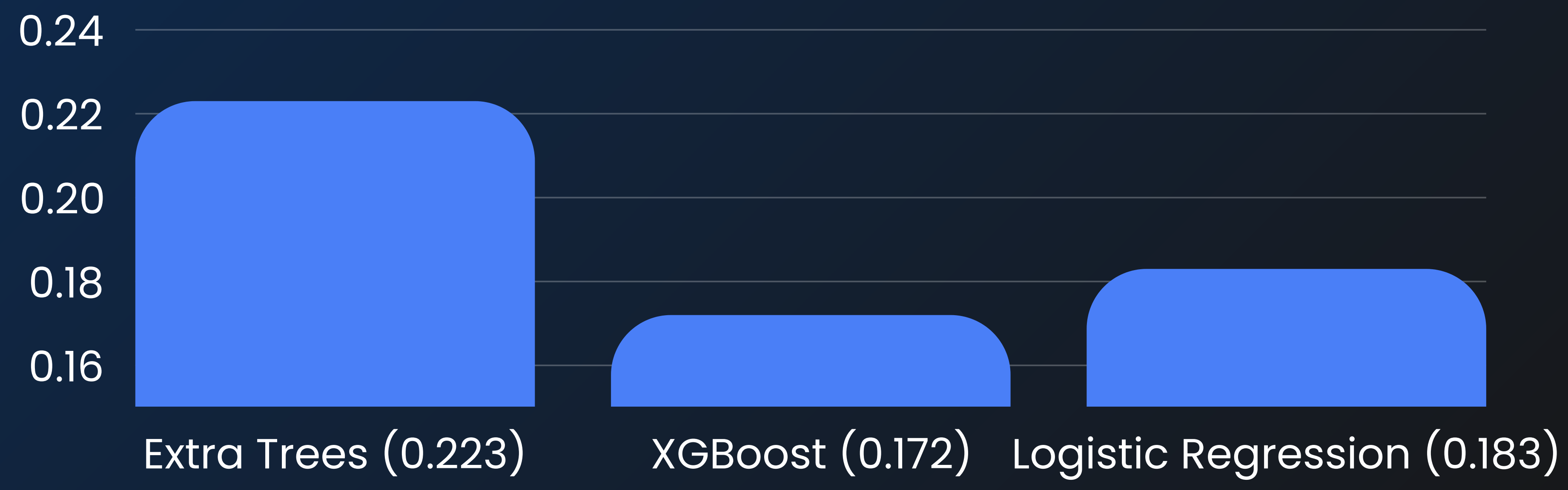
PR-AUC is the right metric here
With only 6.6% positives, standard accuracy is misleading. PR-AUC measures how well the model ranks the rare returners above the majority, directly relevant to campaign efficiency.

3.4× lift over random outreach
The PR-AUC of 0.22 versus the baseline of 0.066 means our ranked list delivers 3.4× more buyers per contact than selecting clients at random.

ROC-AUC (Higher = better)



PR-AUC (Critical for imbalanced dataset)



Proven Precision: Delivering a 3.4x Lift Over Random Targeting

0.70

AUC-ROC Score (benchmark 0.65 - 0.75)

0.22

PR - AUC Score (3.4x above baseline)

3.4x

Lift Over Random (per 100 contacts)

What does AUC-ROC 0.70 mean?

If we randomly select one client who reactivated and one who didn't, the model correctly identifies which is which 70% of the time — versus 50% for random guessing.

This confirms the model has genuine, reliable discriminative power on data it has never seen before.

Validation approach: training on pre-2019 data, tested on Jan–Jun 2020 actual reactivations.

The 3.4× lift — in plain English

If your team contacts 200 clients selected at random from the inactive pool, you would expect to reach approximately 13 genuine buyers (6.6% × 200).

Using our ranked list, contacting the top 200 clients yields approximately 45 genuine buyers — 3.4× more.

Same headcount. Same budget. 3.4× the revenue opportunity.

What Drives Reactivation?

1. Client Risk Category: Financially stable clients buy again. Risk class captures underlying propensity to spend more reliably than any purchase history metric.

2. Industry / Trade Sector: Construction and Crafting follow equipment replacement cycles. Campaigns can be timed to predicted renewal windows.

3. Geographic Region: Certain Italian regions have structurally higher return rates. Local sales teams should prioritise these areas.

4. Max Gap Between Purchases: Some clients naturally buy infrequently. A long silence may be normal behaviour — not a signal of permanent churn.

5. Economic Potential Score: Clients with larger addressable markets have more reasons to re-engage and tend to return at higher rates.

Business Insights

Recency ≠ Comeback

Clients silent 2.1 years score identically to those silent 3+ years.

Do not sort your campaign list by days since last purchase, it adds zero signal. Behavioural patterns are far more predictive than silence duration.

High Spenders Return

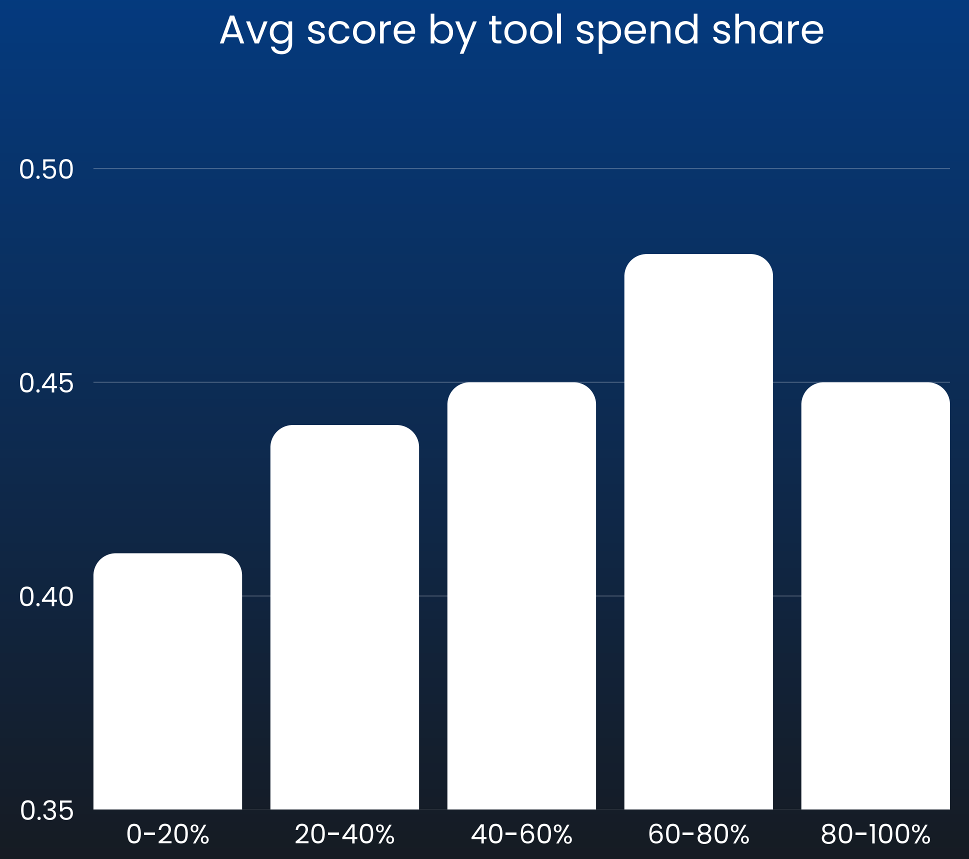
Top spend tier scores 0.49 vs. 0.37 for the lowest tier.

Prioritise clients with high historical spend. Their past investment in the relationship is the strongest leading indicator of willingness to return.

Mixed Buyers Are Best

60–80% tool spend is the sweet spot (score: 0.48).

Target clients who bought both tools and consumables. Pure tool-only buyers score lower, diversity of purchasing signals a broader, stickier relationship.



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The Recall Strategy

Why We Prioritized Recall

In a reactivation campaign, the two types of error have very different costs:

- Calling someone who doesn't buy: costs one email or one sales call, roughly €1–5.
- Missing a client who would have bought: loses a potential €500–2,000 order, permanently.

We tuned the model to minimize missed buyers (high recall), while keeping precision strong enough that the sales team only contacts clients with a genuine mathematical likelihood of purchase.

Threshold	# Clients	Channel	Campaign Type
≥ 0.70	~150	Direct call only	Ultra-lean pilot
≥ 0.55	**1,469**	**Call + direct mail**	**Standard (recommended)**
≥ 0.40	~1,800	Automated email	Aggressive

Confusion Matrix at Threshold 0.55 (Extra Trees)

	Predicted: Not Reactivated	**Predicted: Reactivated**
Actual: Not Reactivated	**2,268 (TN)**	**206 (FP)**
Actual: Reactivated	**116 (FN)**	**59 (TP)**

Strategic Action Plan

All 13,244 clients are now scored. Three campaign options let the company scale effort to available budget.

Tier 1

Standard Campaign

Score ≥ 0.55 | 1,469 clients

Your highest-probability bets. These clients show the strongest combination of past loyalty signals, spending behaviour, and company profile. Contact personally.

Action: Personalised outreach

Channel: Phone calls & direct mail

Priority: Start immediately

Tier 2

Aggressive Campaign

Score 0.40–0.55 | ~1,800 clients

Strong candidates worth targeting if budget allows. Automate with personalised sector messaging, e.g. timing Construction clients to tool replacement cycles.

Action: Automated outreach

Channel: Sector-specific email campaigns

Priority: Month 2

Tier 3

Ultra-Lean Pilot

Score ≥ 0.70 | ~150 clients

A 'sure-bet' subset for a rapid 90-day pilot test. Ideal for measuring campaign effectiveness before scaling to Tier 1 and 2.

Action: High-confidence test

Channel: Direct sales team calls only

Priority: Week 1 pilot

Conclusion & Business Impact

1. Immediate ROI: Contact the 1,469 Tier 1 clients (score ≥ 0.55) via phone and direct mail. Measure conversion rate against the 3.4× lift baseline to validate model performance in the field.

2. Operational Efficiency: The model replaces the current manual process of reviewing inactive client lists. Plug the scoring pipeline directly into the CRM to generate prioritised lists automatically.

3. Scalability: Each quarter, new clients cross the 24-month silence threshold. A quarterly score refresh ensures the next wave of reactivatable clients is captured before they are lost.

4. The Long Term Goal: The company shifts from waiting for clients to return to identifying and reaching them before they are considered permanently churned.

Project At A Glance

13,244	Inactive clients scored
45	Features engineered
0.70	Model AUC-ROC
3.4x	Lift over random
1,469	Tier 1 targets

Thank You!

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